

# 0\_10xHuman\_8CLC\_SoloTE

August 31, 2026

```
[1]: library(Seurat)
library(ggplot2)
library(ggpubr)
library(rcartocolor)
library(pheatmap)
library(dplyr)
library(RColorBrewer)
library(clustree)
getwd()

dataset_id <- "10xHuman_8CLC"
sample_id <- "DUX4"
dir.create("figures_10xHuman_8CLC")
dir.create("data_10xHuman_8CLC")
```

Loading required package: SeuratObject

Loading required package: sp

‘SeuratObject’ was built under R 4.3.3 but the current version is 4.4.1; it is recommended that you reinstall ‘SeuratObject’ as the ABI for R may have changed

‘SeuratObject’ was built with package ‘Matrix’ 1.6.4 but the current version is 1.7.5; it is recommended that you reinstall ‘SeuratObject’ as the ABI for ‘Matrix’ may have changed

Attaching package: ‘SeuratObject’

The following objects are masked from ‘package:base’:

intersect, t

Warning message:

"package 'dplyr' was built under R version 4.4.3"

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

Warning message:

"package 'RColorBrewer' was built under R version 4.4.3"

Loading required package: ggraph

Attaching package: 'ggraph'

The following object is masked from 'package:sp':

geometry

~/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'

Warning message in dir.create("figures\_10xHuman\_8CLC"):

"'figures\_10xHuman\_8CLC' already exists"

Warning message in dir.create("data\_10xHuman\_8CLC"):

"'data\_10xHuman\_8CLC' already exists"

```
[2]: # colorDict = c("Differentiating"="#88535A",  
#               "2CLC"="#EF8264",  
#               "Pluripotent"="#F2CC8F")  
  
# getwd()
```

## 1 Pre

```
[3]: conversionTable <- read.table("annotation/annotation_hg38_conversion_withAge.  
  ↪tsv") # created with annotation_scripts/create_annotations_human.Rmd  
head(conversionTable)
```

|                      |   | chr   | start | end   | strand | sw_score | locusID     | family      | subfan |
|----------------------|---|-------|-------|-------|--------|----------|-------------|-------------|--------|
|                      |   | <chr> | <int> | <int> | <chr>  | <int>    | <chr>       | <chr>       | <chr>  |
| A data.frame: 6 × 16 | 1 | chr1  | 10469 | 11447 | -      | 3612     | TAR1        | telo        | TAR1   |
|                      | 2 | chr1  | 11505 | 11675 | -      | 484      | L1MC5a_dup3 | L1          | L1MC   |
|                      | 3 | chr1  | 11678 | 11780 | -      | 239      | MER5B_dup1  | hAT-Charlie | MER5   |
|                      | 4 | chr1  | 15265 | 15355 | -      | 318      | MIR3_dup11  | MIR         | MIR3   |
|                      | 5 | chr1  | 18907 | 19048 | +      | 239      | L2a_dup29   | L2          | L2a    |
|                      | 6 | chr1  | 19972 | 20405 | +      | 994      | L3_dup3     | CR1         | L3     |

```
[4]: SoloTE_path <- paste0("/mnt/volume_1p5T/results/SoloTEout/", dataset_id, "/",
  ↪ sample_id, "/", sample_id, "_SoloTE_output/", sample_id, "_legacytes_MATRIX")

# get filtered barcodes
STAR_path <- paste0("/mnt/TEresults/snakeake_results/results/STARoutdir/
  ↪ ", dataset_id, "/", sample_id, "/best_Solo.out/Gene")
filteredBarcodes <- read.table(paste0(STAR_path, "/filtered/barcodes.tsv"))$V1
  ↪ # read barcodes selected by STARsolo

# 2CLC sample
legacyTEmatrix <- Seurat::ReadMtx(mtx = paste0(SoloTE_path, "/matrix.mtx"),
  cells = paste0(SoloTE_path, "/barcodes.tsv"),
  features = paste0(SoloTE_path, "/features.tsv"))
  ↪ # read matrix

legacyTEmatrix <- legacyTEmatrix[,filteredBarcodes]

# select TEs
TEs <- grep("SoloTE", rownames(legacyTEmatrix), value = T)
locusTEs <- grep("chr", TEs, value = T)
# subset the matrix keeping only TEs
TEmatrix <- legacyTEmatrix[locusTEs,]
# remove "SoloTE" from the name of the TEs

rownames(TEmatrix) <- gsub("SoloTE\\|", "", rownames(TEmatrix))

# rownames(TEmatrix) <- gsub("\\|", "-", rownames(TEmatrix))
# rownames(TEmatrix) <- gsub("\\_", "-", rownames(TEmatrix))
# rownames(TEmatrix) <- gsub("\\|?", "", rownames(TEmatrix))

# table(rownames(TEmatrix) %in% conversionTable$soloteID)
# # transform into Stellarscope IDs
# rownames(TEmatrix) <-
  ↪ conversionTable$stellarscopeID[match(rownames(TEmatrix),
  ↪ conversionTable$soloteID)]

nCells <- ncol(TEmatrix)
thrMinCells <- round(nCells * 0.02)
```

```
# create Seurat object with shallow filtering of TEs expressed in at least 50
↳ cells and cells expressing at least 50 TEs
objTE <- Seurat::CreateSeuratObject(TEmatrix, project = "SoloTE",
  min.cells = thrMinCells, min.features = 0)
objTE
thrMinCells
```

Warning message:

"Feature names cannot have underscores ('\_'), replacing with dashes ('-')"

Warning message:

"Feature names cannot have pipe characters ('|'), replacing with dashes ('-')"

An object of class Seurat

365333 features across 125 samples within 1 assay

Active assay: RNA (365333 features, 0 variable features)

1 layer present: counts

2

```
[5]: objTE_SoloTE <- objTE#[,filteredBarcodes] # keep only filtered barcodes
objTE_SoloTE
```

An object of class Seurat

365333 features across 125 samples within 1 assay

Active assay: RNA (365333 features, 0 variable features)

1 layer present: counts

```
[6]: # select genes
genes <- setdiff(rownames(legacyTEmatrix), TEs)
# subset the matrix keeping only genes
Genematrix <- legacyTEmatrix[genes,]
# create Seurat object with shallow filtering of genes expressed in at least 50
↳ cells and cells expressing at least 100 genes
Gene <- Seurat::CreateSeuratObject(Genematrix, project = "8CLC",
  min.cells = thrMinCells, min.features = 100)

objGenes_SoloTE <- Gene #[,filteredBarcodes] # keep only filtered barcodes
objGenes_SoloTE
```

An object of class Seurat

15771 features across 125 samples within 1 assay

Active assay: RNA (15771 features, 0 variable features)

1 layer present: counts

```
[7]: # QC TEs

options(repr.plot.width=7, repr.plot.height=6)
```

```
objTE_SoloTE@meta.data$nCount_TE <- objTE_SoloTE@meta.data$nCount_RNA
objTE_SoloTE@meta.data$nFeature_TE <- objTE_SoloTE@meta.data$nFeature_RNA

# Visualize QC metrics as a violin plot
VlnPlot(objTE_SoloTE, features = c("nCount_RNA", "nFeature_RNA"), ncol = 2,
        pt.size = 0, alpha = 0.5) + geom_hline(yintercept = 400)
summary(objTE_SoloTE$nCount_RNA)
summary(objTE_SoloTE$nFeature_RNA)
```

Warning message:

"Default search for "data" layer in "RNA" assay yielded no results; utilizing "counts" layer instead."

Warning message:

"The `slot` argument of `FetchData()` is deprecated as of SeuratObject 5.0.0.

Please use the `layer` argument instead.

The deprecated feature was likely used in the [Seurat](#) package.

Please report the issue at

[<https://github.com/satijalab/seurat/issues>.](https://github.com/satijalab/seurat/issues)"

Warning message:

"`PackageCheck()` was deprecated in SeuratObject 5.0.0.

Please use `rlang::check\_installed()` instead.

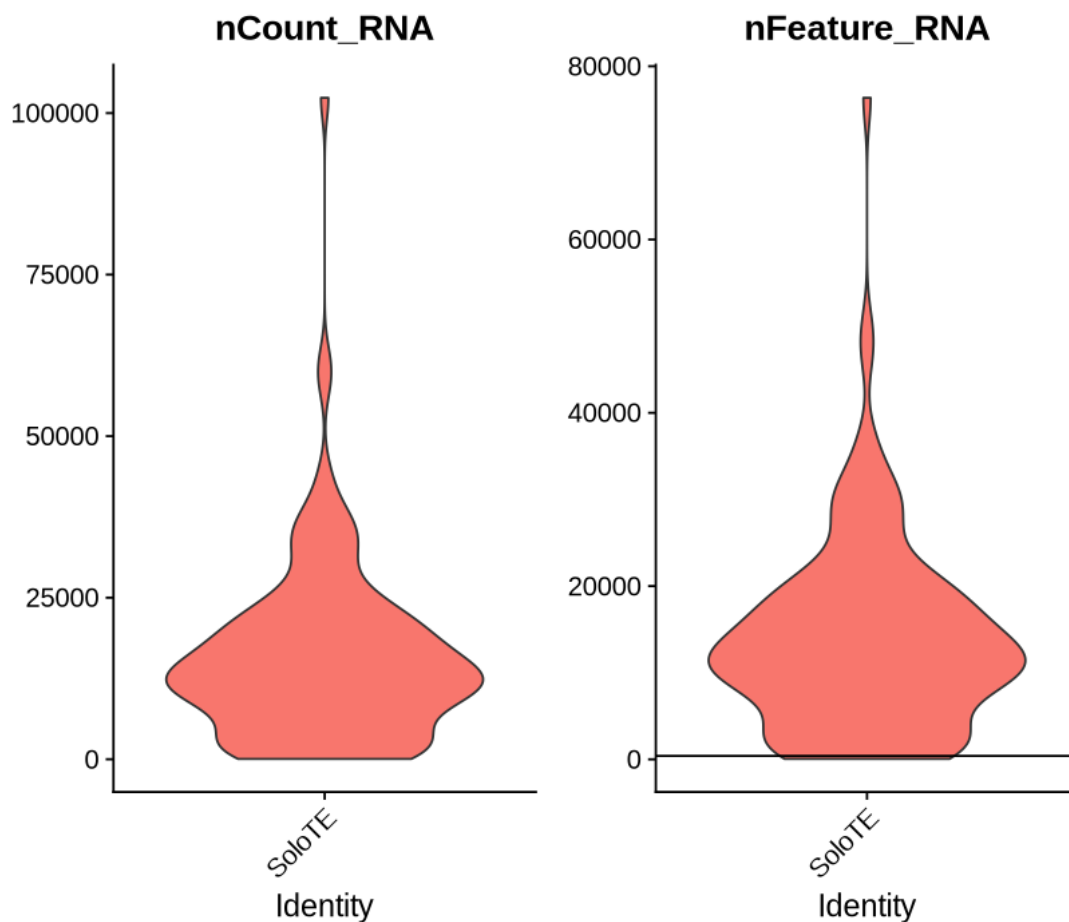
The deprecated feature was likely used in the [Seurat](#) package.

Please report the issue at

[<https://github.com/satijalab/seurat/issues>.](https://github.com/satijalab/seurat/issues)"

| Min. | 1st Qu. | Median | Mean  | 3rd Qu. | Max.   |
|------|---------|--------|-------|---------|--------|
| 50   | 7676    | 12890  | 15809 | 20815   | 102358 |

| Min. | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|------|---------|--------|-------|---------|-------|
| 49   | 7087    | 11817  | 13888 | 18791   | 76359 |



```
[8]: # QC genes
options(repr.plot.width=8, repr.plot.height=5)

VlnPlot(objGenes_SoloTE, features = c( "nCount_RNA", "nFeature_RNA" ), ncol = 2,
      pt.size = 0.05, alpha = 0.5) +
  geom_hline(yintercept = 2000)

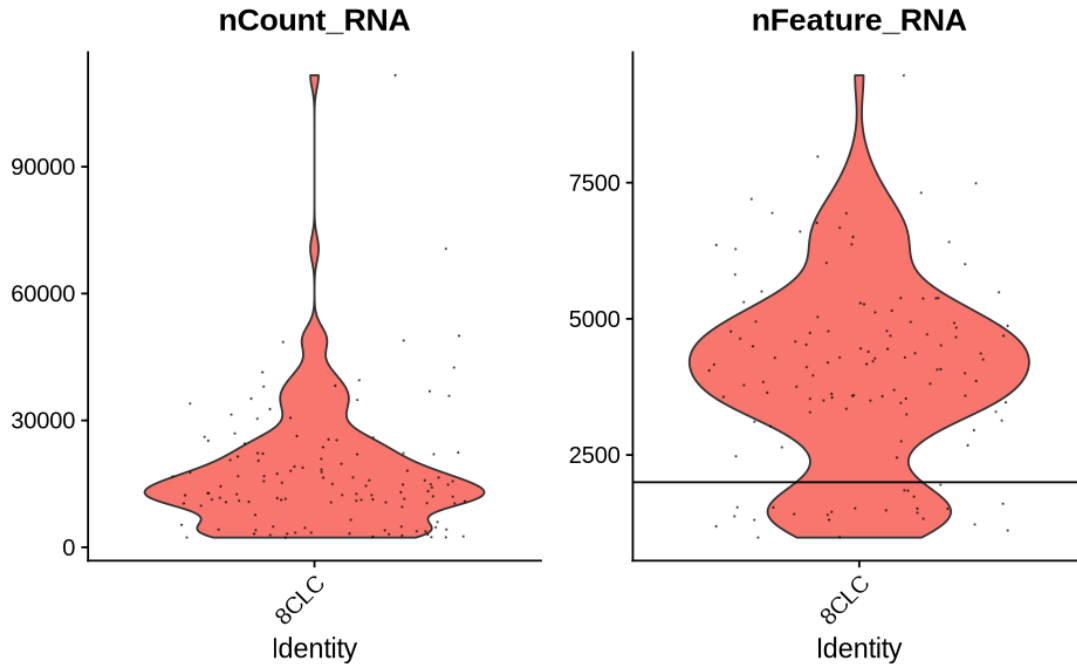
summary(objGenes_SoloTE$nFeature_RNA)
summary(objGenes_SoloTE$nCount_RNA)
```

Warning message:

"Default search for "data" layer in "RNA" assay yielded no results; utilizing "counts" layer instead."

|      |         |        |      |         |      |
|------|---------|--------|------|---------|------|
| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 983  | 3110    | 4115   | 4014 | 4941    | 9471 |
| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |

2290    10345    14525    17510    22135    111612



## 2 Genes

```
[9]: gc()

objGenes_SoloTE <- NormalizeData(objGenes_SoloTE, normalization.method = "LogNormalize",
  scale.factor = 10000)

objGenes_SoloTE <- FindVariableFeatures(objGenes_SoloTE, selection.method = "vst",
  nfeatures = 4000)

# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(objGenes_SoloTE), 10)

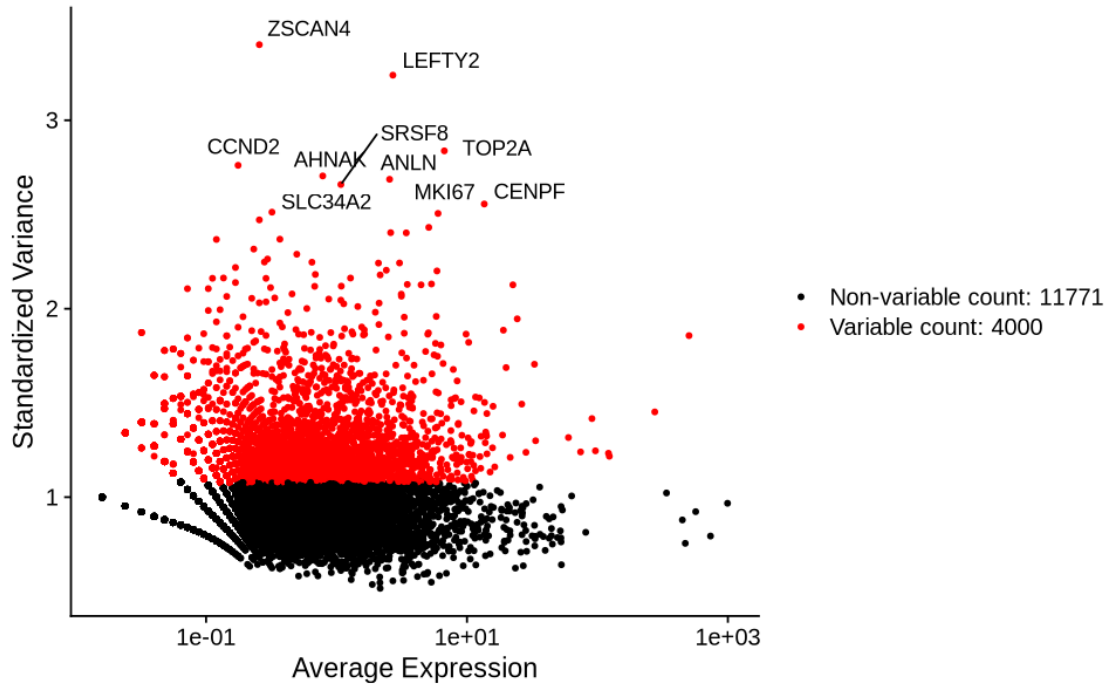
# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objGenes_SoloTE)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2
```

|                                    |        | used      | (Mb)   | gc trigger | (Mb)   | max used  | (Mb)   |
|------------------------------------|--------|-----------|--------|------------|--------|-----------|--------|
| A matrix: $2 \times 6$ of type dbl | Ncells | 27906108  | 1490.4 | 64040821   | 3420.2 | 36651804  | 1957.5 |
|                                    | Vcells | 233489031 | 1781.4 | 423292210  | 3229.5 | 422650634 | 3224.6 |

Normalizing layer: counts

Finding variable features for layer counts

When using `repel`, set `xnudge` and `ynudge` to 0 for optimal results



```
[10]: gc()
all.genes <- rownames(objGenes_SoloTE)
objGenes_SoloTE <- ScaleData(objGenes_SoloTE) # on hvgs

objGenes_SoloTE <- RunPCA(objGenes_SoloTE, features = VariableFeatures(object =
  ↪objGenes_SoloTE))

DimPlot(objGenes_SoloTE, reduction = "pca") + NoLegend()

ElbowPlot(objGenes_SoloTE)
```

|                             |        | used      | (Mb)   | gc trigger | (Mb)   | max used  | (Mb)   |
|-----------------------------|--------|-----------|--------|------------|--------|-----------|--------|
| A matrix: 2 × 6 of type dbl | Ncells | 27941093  | 1492.3 | 64040821   | 3420.2 | 36651804  | 1957.5 |
|                             | Vcells | 234454094 | 1788.8 | 423292210  | 3229.5 | 422650634 | 3224.6 |

Centering and scaling data matrix

PC\_1

Positive: MALAT1, KCNQ10T1, DDX17, ZNF208, PDS5A, MYSM1, RPSA, PROM1, CENPC,



NEAT1

SLC25A16, AL390728.6, TNPO1, SNHG14, EP300, WSB1, ZMYM2, ZNF695,  
MED13, BPTF

MDN1, RABGGTB, TMP0, PAXBP1, WAPL, PHIP, ZNF292, REST, HNRNPH1, TULP4  
Negative: COX6C, POLR2J, MT-ND5, MT-ND2, AEBP1, C9orf135, MT-RNR1, C2orf16, MT-  
ND3, CD80

ATAD3C, RIOX1, ACOT7, HBA2, USP40, S1PR3, AC027559.1, HLA-B,  
AC105760.2, AL445222.1

NDUFAF3, SEPT12, MMD, RASGRP2, DNAJC4, GGH, AC040169.1, CUEDC1,  
RNASEH2C, HIST1H2AA

PC\_ 2

Positive: SLC34A2, TRIM43, AC126177.5, S100A6, EFNA1, CCNA1, H3.Y, TPRX1,  
AC106872.11, TRIM53CP

POU3F1, FZD7, ALPP, WDR47, UNC5B-AS1, MBD3L2, RFPL2, JUNB, MBD3L2B,  
GDF15

AL513321.2, MBD3L5, LEUTX, RFPL1, TRIB1, TRIM49B, IL11, PNP, TC2N,  
RFPL4B

Negative: MT-ND3, MT-ND5, MT-RNR1, MT-ND2, MT-ND1, MT-ND4L, AASS, MT-ATP8,  
GDF3, EPCAM

MT2A, TET1, SLC16A10, HSP90B1, ZNF676, PSMD12, FZD5, RAB3B, SMC2,  
MFSD4B

GLDC, MT1F, CANX, SNHG14, LGR4, NUCKS1, BICD1, ASPM, SLC44A1, LBR

PC\_ 3

Positive: MT-ND2, NAP1L1, NLRP2, HNRNPK, HSPA9, HSP90AB1, MT-ND4L, HSPD1, EEF2,  
CALR

ACSL3, MT-ND3, TUBA1C, SPARC, MT2A, LCP1, PDIA6, CANX, MT-ND5, MT-ND1  
ATP1B3, APLP2, EDF1, HSP90AA1, MT-ATP8, MT-RNR1, DSP, UBB, PDIA4,

EPCAM

Negative: SEC31B, AL117329.1, AC114939.1, MAGEB1, SOD3, AL121845.1, CRYZL2P,  
TTC39C, RARB, RPS15AP29

FAM174B, AC026124.2, AC099654.14, PDE4D, AL138688.2, AL451064.2,  
TMC5, MYH7B, FSD1L, SLFN13

KCTD13, AC093525.4, CYFIP2, DIABLO, ZNF155, ARHGAP26, AC130324.3,  
PLSCR4, B3GALT6, AC007996.1

PC\_ 4

Positive: TRIM53CP, MBD3L2, MBD3L2B, AC106872.11, RFPL2, TC2N, TPRX1, RFPL1,  
CCNA1, LEUTX

AC126177.5, MBD3L5, TRIM49B, AL513321.2, MYC, PRAMEF1, UNC80, KLF18,  
AC025171.5, SOX9

OR10G3, AP002383.3, IMPDH1, GRM2, H3.Y, HBA1, KLF10, RFPL4B, LACC1,  
ZSCAN4

Negative: AC023509.2, CCDC168, TENT5C, CPM, AC132872.4, RASGRP1, AP002433.2,  
ZNF688, NKD2, C6orf48

AL391095.3, LINC00471, LINC00475, LARP4P, AL109976.1, CYP27B1,  
ITPRID2, CCT6P3, ATP1B2, ZNF14

TEX30, DEPDC7, MFSD2A, LHFPL2, RNF103, AL731563.3, LINC00324, FKBP9,  
TAF6L, HECA

PC\_ 5

Positive: RBMX2P3, LBHD1, DRC3, TSIX, KLHL22, MFSD1P1, AC006206.1, PGAM1P6, AC011379.1, USP2

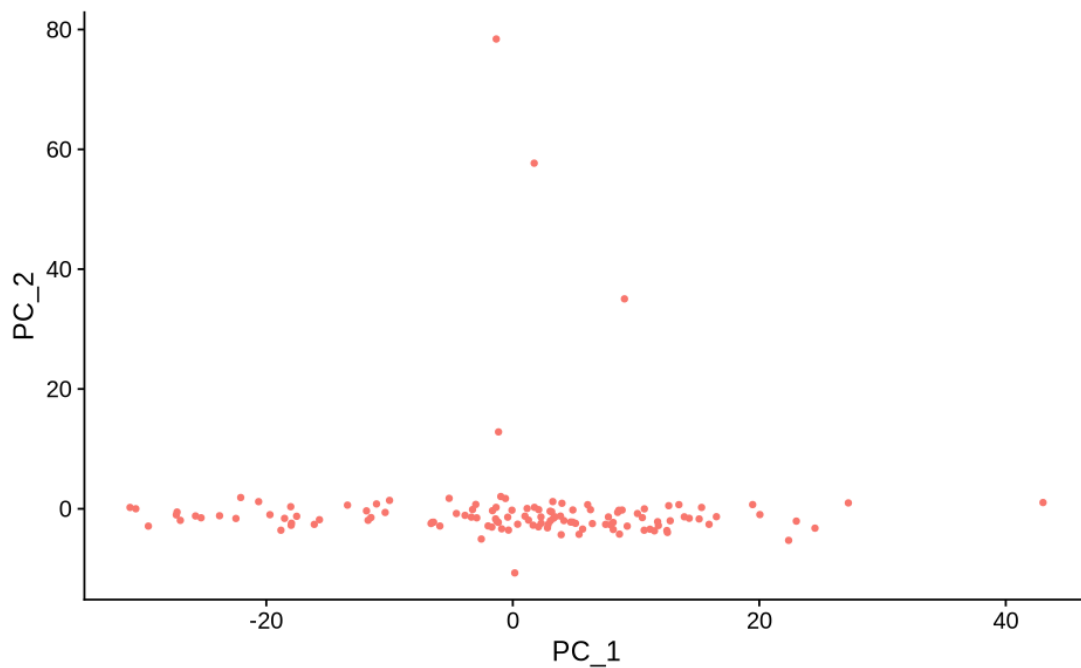
AL513550.1, AC104073.4, INE1, CHRNE, PHYH, SMCR8, ITPRIP, WWOX, MCOLN3, METTL4

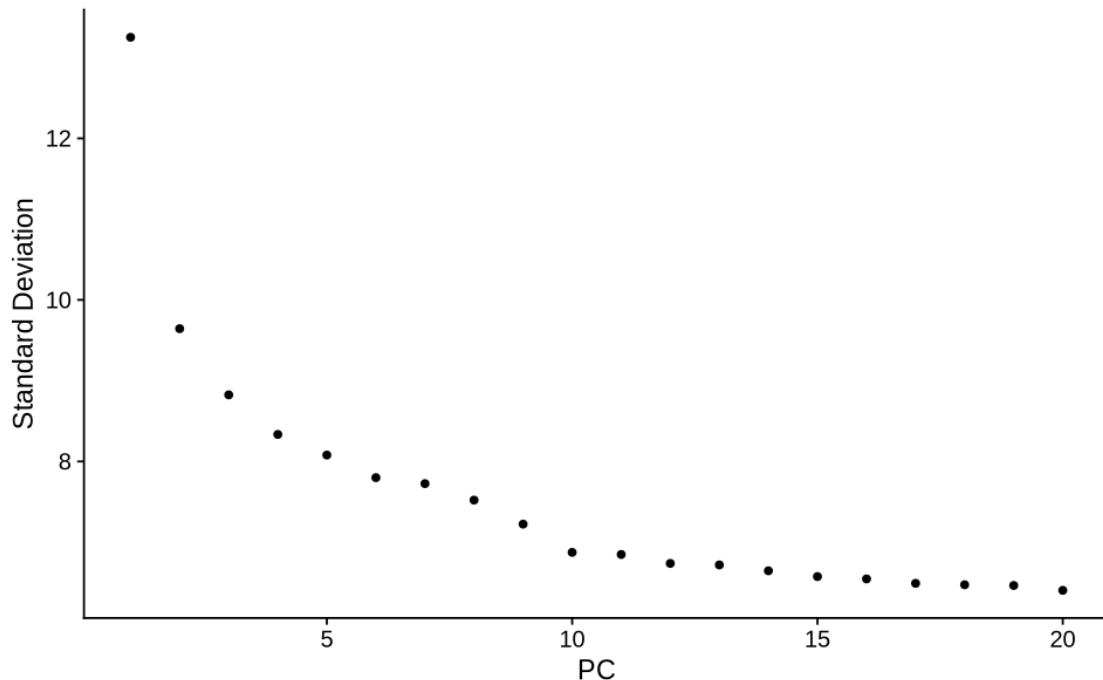
EPS8L2, LIPT2, ISL1, AC129502.1, EFCAB13, AC134407.3, RHBDF1, ZKSCAN2-DT, AC108860.1, AC092329.4

Negative: VIRMA, RPL35, ALOX12-AS1, SEMA4C, UMAD1, ATP5MG, FABP5, PLBD1, SEC31A, SMC2

RPS15AP29, NOL7, CCN2, ANP32A, HMGB1, NCL, TRIP13, FAM174B, NAA16, PHACTR4

PRKAR2A, RPL6, AL117329.1, AL451064.2, PDE4D, AC026124.2, RANBP2, CRYZL2P, LCA5, FSD1L





```
[11]: objGenes_SoloTE <- FindNeighbors(objGenes_SoloTE, dims = 1:10, k.param = 20)
      objGenes_SoloTE <- FindClusters(objGenes_SoloTE, resolution = 1, algorithm = 4)
      objGenes_SoloTE <- RunUMAP(objGenes_SoloTE, dims = 1:10)
      DimPlot(objGenes_SoloTE, reduction = "umap")
```

Computing nearest neighbor graph

Computing SNN

Warning message in RunLeiden(object = object, method = method, partition.type = "RBConfigurationVertexPartition", :

"`random.seed` must be greater than 0 for leiden clustering, resetting  
`random.seed` to 1."

Warning message:

"The default method for RunUMAP has changed from calling Python UMAP via reticulate to the R-native UWOT using the cosine metric  
To use Python UMAP via reticulate, set umap.method to 'umap-learn' and metric to 'correlation'

This message will be shown once per session"

15:29:30 UMAP embedding parameters a = 0.9922 b = 1.112

15:29:30 Read 125 rows and found 10 numeric columns

15:29:30 Using Annoy for neighbor search, n\_neighbors = 30

```
15:29:30 Building Annoy index with metric = cosine, n_trees = 50
```

0% 10 20 30 40 50 60 70 80 90 100%

[-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|

\*\*\*\*\*

\*  
\*  
\*  
\*  
\*  
\*  
\*  
\*  
|

15:29:30 Writing NN index file to temp file  
/mnt/TEdataStorage\_2T/tmp/RtmpjoyMJm/file376f76187240a

15:29:30 Searching Annoy index using 1 thread, search\_k = 3000

15:29:30 Annoy recall = 100%

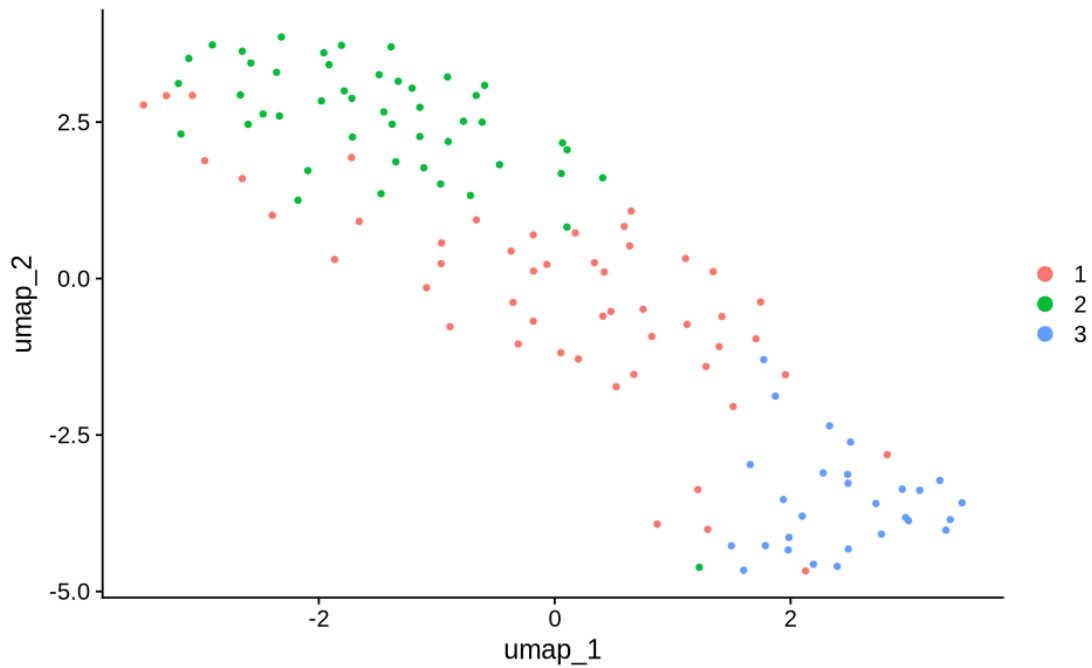
15:29:35 Commencing smooth kNN distance calibration using 1 thread  
with target n\_neighbors = 30

15:29:45 Initializing from normalized Laplacian + noise (using RSpectra)

15:29:45 Commencing optimization for 500 epochs, with 4554 positive edges

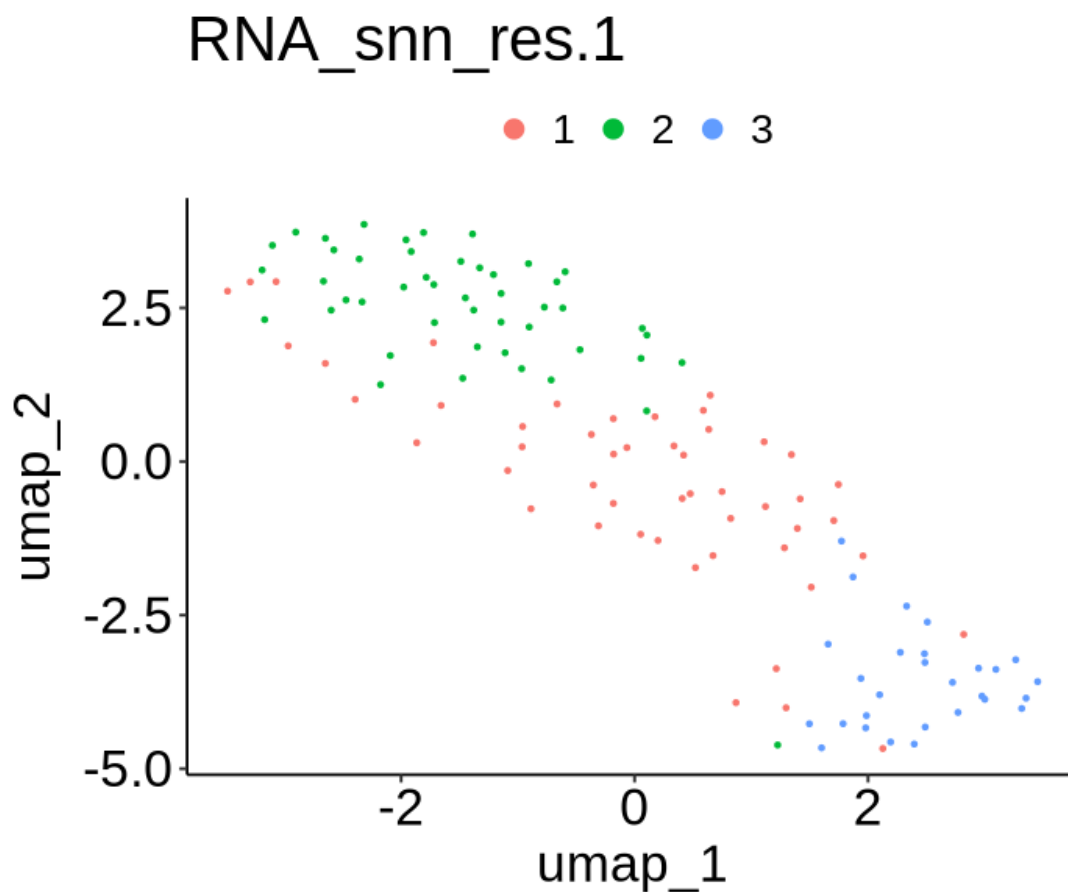
15:29:45 Using rng type: pcg

15:29:50 Optimization finished



```
[12]: options(repr.plot.width=6, repr.plot.height=5)

DimPlot(objGenes_SoloTE, reduction = "umap", group.by="RNA_snn_res.1",
  #cols=colorDict,
  shuffle=T, pt.size = 0.5) +
  theme_pubr() +
  theme(text=element_text(size=20))
ggsave(paste0("figures_",dataset_id,"/umap_genes_clusters.pdf"), device = "pdf", width=6, height=5)
```



### 3 TEs

```
[17]: ### Normalize

objTE_SoloTE <- NormalizeData(objTE_SoloTE, normalization.method = "
  ↪LogNormalize", scale.factor = 10000)

#saveRDS(objTE_SoloTE, file=paste0("data_",dataset_id,"/objTE_SoloTE_beforeHVG.
  ↪RDS"))
```

Normalizing layer: counts

```
[18]: objTE_SoloTE <- FindVariableFeatures(objTE_SoloTE, selection.method = "vst",
  ↪nfeatures = 4000)

# # Identify the 10 most highly variable genes
```

```

top10 <- head(VariableFeatures(objTE_SoloTE), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objTE_SoloTE)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2

```

Finding variable features for layer counts

```

Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"at -1.8103"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"radius 0.00020677"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"all data on boundary of neighborhood. make span bigger"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"pseudoinverse used at -1.8103"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"neighborhood radius 0.01438"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"reciprocal condition number 1"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"There are other near singularities as well. 0.01561"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"zero-width neighborhood. make span bigger"
Rasterizing points since number of points exceeds 100,000.
To disable this behavior set `raster=FALSE`

```

When using repel, set xnudge and ynudge to 0 for optimal results

```

Warning message:
"Removed 192863 rows containing missing values or values outside the
scale range
(`geom_scattermore()`)."

```





3. 'chr10-100942047-100942492-MER21C:ERVL:LTR-22.5-' 4. 'chr10-100942956-100943284-MER21C:ERVL:LTR-  
5. 'chr10-101054661-101055130-LTR16C:ERVL:LTR-30.7-' 6. 'chr10-101368005-101368435-MSTB1:ERVL-MaLR:L  
7. 'chr10-101392848-101393474-MER21A:ERVL:LTR-15.1-+' 8. 'chr10-101393640-101393744-MER21A:ERVL:LTR  
9. 'chr10-101480342-101480635-MLT1A0:ERVL-MaLR:LTR-19.7-' 10. 'chr10-101545409-101545775-MLT1B:ERVL  
11. 'chr10-101576646-101576835-MLT1D:ERVL-MaLR:LTR-22.1-' 12. 'chr10-101729635-101729855-THE1C:ERVL  
13. 'chr10-101977560-101977988-MSTC:ERVL-MaLR:LTR-15.9-+' 14. 'chr10-101983547-101984186-MER21B:ERV  
15. 'chr10-101987525-101987858-MLT1F:ERVL-MaLR:LTR-21.8-+'  
16. 'chr10-102287194-102287231-MLT1F:ERVL-MaLR:LTR-5.4-' 17. 'chr10-102287251-102287329-MLT1F:ERVL-M  
18. 'chr10-102287488-102287885-MLT1F:ERVL-MaLR:LTR-20.0-' 19. 'chr10-102322959-102323430-MLT1H1:ERV  
20. 'chr10-102383871-102384524-MER21B:ERVL:LTR-16.4-+' 21. 'chr10-102442633-102442976-LTR42:ERVL:LTR  
22. 'chr10-102455857-102456339-MLT1D:ERVL-MaLR:LTR-20.2-+'  
23. 'chr10-102456581-102456807-MLT1K:ERVL-MaLR:LTR-33.6-+'  
24. 'chr10-102456844-102456938-MLT1K:ERVL-MaLR:LTR-29.5-+'  
25. 'chr10-102574019-102574189-MLT1F2:ERVL-MaLR:LTR-21.3-+'  
26. 'chr10-102720864-102721218-MLT2C1:ERVL:LTR-21.7-+' 27. 'chr10-102804536-102804924-MLT1I:ERVL-MaL  
28. 'chr10-102928237-102928304-MLT1A1:ERVL-MaLR:LTR-19.3-+'  
29. 'chr10-103133634-103135208-THE1B-int:ERVL-MaLR:LTR-9.3-+'  
30. 'chr10-103258768-103259112-THE1B:ERVL-MaLR:LTR-13.0-+'  
31. 'chr10-103308570-103308590-MSTA-int:ERVL-MaLR:LTR-9.7-+'  
32. 'chr10-103429426-103429596-MLT1B:ERVL-MaLR:LTR-14.0-' 33. 'chr10-103608243-103608415-MLT1C:ERVL  
34. 'chr10-103687046-103687368-MLT2C1:ERVL:LTR-26.8-+' 35. 'chr10-103720020-103720248-MLT1A1:ERVL-Ma  
36. 'chr10-103983439-103984127-LTR50:ERVL:LTR-21.3-+' 37. 'chr10-104262206-104262428-MLT1N2:ERVL-MaL  
38. 'chr10-104279867-104279889-MLT1J:ERVL-MaLR:LTR-31.1-' 39. 'chr10-104280192-104280591-MLT1J:ERVL-  
40. 'chr10-1066717-1067070-THE1B:ERVL-MaLR:LTR-9.9-' 41. 'chr10-110023180-110023464-MLT1A0:ERVL-MaL  
42. 'chr10-110038280-110038401-MLT1J2:ERVL-MaLR:LTR-24.4-'  
43. 'chr10-110049906-110050213-THE1B-int:ERVL-MaLR:LTR-9.6-+'  
44. 'chr10-110087792-110088241-MLT1C:ERVL-MaLR:LTR-19.1-' 45. 'chr10-112589166-112589539-THE1B:ERVL  
46. 'chr10-112591737-112592355-MLT1E2:ERVL-MaLR:LTR-23.1-+'  
47. 'chr10-112705218-112705353-MLT1N2:ERVL-MaLR:LTR-27.4-'  
48. 'chr10-112705617-112705687-LTR67B:ERVL:LTR-26.8-' 49. 'chr10-112809992-112810120-MLT1K:ERVL-MaL  
50. 'chr10-112810421-112810728-MLT1K:ERVL-MaLR:LTR-30.5-'

```
[20]: objTE_SoloTE <- RunPCA(objTE_SoloTE, features = VariableFeatures(object = objTE_SoloTE))

DimPlot(objTE_SoloTE, reduction = "pca") + NoLegend()

ElbowPlot(objTE_SoloTE)
```

PC\_ 1  
Positive: chr13-79352619-79352851-MIRc:MIR:SINE-24.7--,  
chr12-70505981-70506273-AluJr:Alu:SINE-17.8-+,  
chr1-32654091-32654389-AluJb:Alu:SINE-15.1-+,  
chr18-46096219-46096517-AluSp:Alu:SINE-7.0-+,  
chr14-34600127-34600423-AluSz6:Alu:SINE-15.2--,  
chr16-71804735-71805012-AluJb:Alu:SINE-15.5-+,  
chr12-76480805-76481262-MLT1D:ERVL-MaLR:LTR-18.9-+,

chr14-39162856-39163069-L2a:L2:LINE-29.0-+,  
chr5-109745972-109746274-AluSz6:Alu:SINE-14.5--,  
chr14-45088525-45088665-MIRb:MIR:SINE-32.4-+  
chr19-53650784-53651296-MER57-int:ERV1:LTR-13.5-+,  
chr3-177078311-177078605-AluJo:Alu:SINE-14.6-+,  
chr19-53658600-53658909-AluSx1:Alu:SINE-10.7-+,  
chr2-10350600-10351475-L1MB8:L1:LINE-14.8-+,  
chr12-112428748-112429046-AluSp:Alu:SINE-7.4--,  
chr20-18515951-18516252-L2a:L2:LINE-26.6-+,  
chr10-118660589-118662884-L1PBa:L1:LINE-8.3--,  
chr10-93029899-93030207-AluSp:Alu:SINE-9.1-- , chr12-8788669-8793938-HERV9NC-  
int:ERV1:LTR-4.2-+ , chr15-34950079-34950387-AluSc:Alu:SINE-9.7--  
chr15-81832145-81832235-L1PB3:L1:LINE-13.2-+ ,  
chr13-26317215-26319185-L1MB3:L1:LINE-19.0-+ ,  
chr14-45234936-45235222-AluSx:Alu:SINE-13.2-+ ,  
chr3-52592123-52592430-AluSx:Alu:SINE-9.7-- ,  
chr22-17391427-17391614-MIRc:MIR:SINE-27.0-+ ,  
chr18-70170183-70171274-L2a:L2:LINE-29.0-- ,  
chr3-197169754-197170910-L1PA13:L1:LINE-11.5-- ,  
chr19-21477254-21477533-AluSp:Alu:SINE-10.7-+ ,  
chr5-150380727-150381031-AluSx1:Alu:SINE-7.5-+ ,  
chr5-37699666-37699974-AluSz:Alu:SINE-11.7-+  
Negative: chr11-109154301-109155116-L1MEf:L1:LINE-27.3-- ,  
chr12-59569475-59569921-MLT2A1:ERVL:LTR-9.1-- ,  
chr11-49029779-49030051-Tigger3a:TcMar-Tigger:DNA-16.5-+ ,  
chr13-38579023-38579320-AluSz:Alu:SINE-13.8-- ,  
chr13-36433549-36433840-AluSq2:Alu:SINE-6.5-- ,  
chr11-91232981-91233333-THE1B:ERVL-MaLR:LTR-14.7-- ,  
chr10-65140755-65141709-L1MA10:L1:LINE-20.6-+ ,  
chr13-38569709-38573029-L1PA16:L1:LINE-14.4-+ ,  
chr5-121384255-121384558-THE1C:ERVL-MaLR:LTR-12.0-- ,  
chr9-31677328-31677682-L1MEc:L1:LINE-21.9--  
chr13-32866497-32866807-AluJo:Alu:SINE-19.3-+ ,  
chr11-109152513-109152808-AluSx1:Alu:SINE-10.1-- ,  
chr8-77854747-77856471-L1MC1:L1:LINE-20.5-- ,  
chr21-23032113-23033697-L1PA14:L1:LINE-12.8-- ,  
chrX-111666325-111666468-MIRc:MIR:SINE-33.8-+ ,  
chr13-38579323-38579420-Tigger20a:TcMar-Tigger:DNA-32.6-- ,  
chr16-24556016-24556251-MIR:MIR:SINE-22.1-+ , chr13-38579425-38579765-MSTB1:ERVL-  
MaLR:LTR-14.4-+ , chr1-57599981-57600302-L2b:L2:LINE-28.0-+ ,  
chr19-53285641-53285768-MIR:MIR:SINE-27.8-+  
chr1-12792582-12792820-AluSz:Alu:SINE-20.1-- ,  
chr21-44494305-44494600-AluSg:Alu:SINE-11.8-- ,  
chr11-91232774-91232980-L2a:L2:LINE-24.3-- ,  
chr11-49030052-49030366-AluSz6:Alu:SINE-15.6-+ ,  
chrX-153779329-153779629-AluSx:Alu:SINE-9.6-- ,  
chr9-89341579-89342940-L1MB4:L1:LINE-19.9-+ ,  
chr7-152537253-152537556-AluSx1:Alu:SINE-12.8-- ,

chr3-138172648-138173216-L1MC5a:L1:LINE-25.9-+,  
chr11-109136730-109140066-L1M1:L1:LINE-10.7--,  
chr3-104100291-104100712-L1PREC2:L1:LINE-10.8--  
PC\_ 2

Positive: chr4-46806434-46813159-L1MA9:L1:LINE-24.8--,  
chr6-16266133-16266520-LTR12:ERV1:LTR-18.7-+,  
chr4-34134167-34138437-L1PA16:L1:LINE-16.2--,  
chr2-42413951-42415100-L1MEi:L1:LINE-25.9-- , chr13-69281000-69281387-MLT1B:ERVL-  
MaLR:LTR-21.4-- , chr15-57433665-57435259-HAL1:L1:LINE-28.0-+ ,  
chr13-69385653-69387806-L1M1:L1:LINE-24.2-+ ,  
chr9-131398621-131398686-L2a:L2:LINE-28.8-- ,  
chr6-121150010-121154449-L1PA7:L1:LINE-11.5-- ,  
chrX-23033867-23034462-L1MEf:L1:LINE-24.7--  
chrX-23534970-23535258-L1MC5a:L1:LINE-28.2-+ ,  
chr2-195760053-195766167-L1MA3:L1:LINE-11.4-+ ,  
chr6-160883402-160884426-PRIMA4-int:ERV1:LTR-13.2-+ ,  
chr6-73603421-73603719-AluSq2:Alu:SINE-12.7-- ,  
chr12-51052501-51052806-AluSg:Alu:SINE-10.1-+ ,  
chr2-32319006-32319318-AluSp:Alu:SINE-8.9-+ ,  
chr11-39111007-39112294-HSMAR1:TcMar-Mariner:DNA-10.6-+ ,  
chr1-158016463-158016778-Tigger19a:TcMar-Tigger:DNA-30.1-- ,  
chr12-113312897-113313195-AluSx1:Alu:SINE-13.0-- ,  
chr19-20181260-20181564-AluYh3:Alu:SINE-6.6-+  
chr3-186957477-186958658-HAL1:L1:LINE-27.1-+ ,  
chr22-45162882-45163197-AluSx:Alu:SINE-8.9-+ ,  
chr2-218565700-218565766-L2c:L2:LINE-29.0-- ,  
chr13-69358196-69359460-L1PA5:L1:LINE-4.2-+ ,  
chr5-18645999-18648513-L1MA3:L1:LINE-11.8-- ,  
chr15-81648650-81649927-L1PB3:L1:LINE-12.3-- ,  
chr13-69254714-69259046-L1MA2:L1:LINE-16.3-+ ,  
chrX-114442048-114444617-L1M4c:L1:LINE-23.7-- ,  
chr4-37001706-37002035-MER48:ERV1:LTR-9.3-+ ,  
chr19-12143985-12144271-AluSx1:Alu:SINE-13.2-+  
Negative: chr13-36433549-36433840-AluSq2:Alu:SINE-6.5-- ,  
chr5-121384255-121384558-THE1C:ERVL-MaLR:LTR-12.0-- ,  
chr16-24542457-24542745-AluSz6:Alu:SINE-11.4-- ,  
chr1-57599981-57600302-L2b:L2:LINE-28.0-+ ,  
chrX-111666325-111666468-MIRc:MIR:SINE-33.8-+ ,  
chr9-31677328-31677682-L1MEc:L1:LINE-21.9-- ,  
chr13-38569709-38573029-L1PA16:L1:LINE-14.4-+ ,  
chrX-153779329-153779629-AluSx:Alu:SINE-9.6-- ,  
chr11-49030052-49030366-AluSz6:Alu:SINE-15.6-+ ,  
chr13-38579425-38579765-MSTB1:ERVL-MaLR:LTR-14.4-+  
chr3-104100710-104104158-L1PREC2:L1:LINE-20.9-- ,  
chr13-38579023-38579320-AluSz:Alu:SINE-13.8-- ,  
chr11-91232774-91232980-L2a:L2:LINE-24.3-- ,  
chr12-59569475-59569921-MLT2A1:ERVL:LTR-9.1-- ,  
chr13-38579323-38579420-Tigger20a:TcMar-Tigger:DNA-32.6-- ,

chr11-109136730-109140066-L1M1:L1:LINE-10.7--,  
chr11-49029779-49030051-Tigger3a:TcMar-Tigger:DNA-16.5-+,  
chr1-114294733-114297115-HAL1:L1:LINE-26.8-+,  
chr17-82633729-82633867-AluJr:Alu:SINE-18.0-+,  
chr11-109154301-109155116-L1MEf:L1:LINE-27.3--  
chr21-23032113-23033697-L1PA14:L1:LINE-12.8--,  
chr10-65140755-65141709-L1MA10:L1:LINE-20.6-+,  
chr11-91232981-91233333-THE1B:ERVL-MaLR:LTR-14.7--,  
chr13-106847242-106847618-THE1D:ERVL-MaLR:LTR-13.6-+,  
chr2-183191847-183192033-MLT1D:ERVL-MaLR:LTR-14.8-+,  
chr16-24538075-24538325-L2a:L2:LINE-29.3--,  
chr13-32866497-32866807-AluJo:Alu:SINE-19.3-+,  
chr11-109152513-109152808-AluSx1:Alu:SINE-10.1--,  
chr17-82633868-82634174-AluSx1:Alu:SINE-9.4-+,  
chr4-38669611-38669923-AluSq:Alu:SINE-8.0-+  
PC\_ 3  
Positive: chr1-42711804-42712677-L1MA4A:L1:LINE-19.6--,  
chr3-50568446-50568754-AluSq2:Alu:SINE-8.1-+,  
chr8-26404515-26404807-AluSz:Alu:SINE-13.3--,  
chr3-49019480-49019773-AluSp:Alu:SINE-5.8--,  
chr6-17687833-17688147-AluY:Alu:SINE-8.2-+,  
chr12-6718792-6719092-AluY:Alu:SINE-9.3-+,  
chr17-18215717-18216009-AluSz6:Alu:SINE-13.3-+,  
chr5-132949530-132949874-AluYm1:Alu:SINE-10.1-+,  
chr3-53253297-53253488-AluJo:Alu:SINE-17.1--,  
chr14-22927085-22927405-AluSx3:Alu:SINE-13.7--  
chr2-231446048-231446465-L1ME2:L1:LINE-22.8--,  
chr12-14795450-14795759-AluSx4:Alu:SINE-12.9-+,  
chr1-150380952-150381144-L2a:L2:LINE-27.6-+,  
chr5-37394392-37394826-L2a:L2:LINE-32.2--,  
chr3-53252842-53253141-AluSq2:Alu:SINE-11.3--,  
chr13-19738304-19738521-MER2:TcMar-Tigger:DNA-16.2--,  
chr20-34364595-34364911-AluSx3:Alu:SINE-5.7-+,  
chr7-65981535-65981701-AluJr:Alu:SINE-13.0-+,  
chr8-33495686-33495958-AluSg4:Alu:SINE-9.8--,  
chr2-182981627-182981930-AluSx1:Alu:SINE-12.2-+  
chr7-139153461-139156603-L1MB8:L1:LINE-16.9--,  
chr19-39444961-39445096-FLAM-C:Alu:SINE-15.4-+,  
chr17-43503218-43503513-AluSq2:Alu:SINE-11.5-+,  
chr20-56479525-56481093-Tigger1:TcMar-Tigger:DNA-13.0--,  
chr6-17605073-17605379-AluSz6:Alu:SINE-10.8--,  
chr10-69142183-69142493-AluJb:Alu:SINE-17.4--,  
chr1-179905673-179905965-AluSx1:Alu:SINE-13.7-+,  
chr19-21311292-21311540-L1MC4:L1:LINE-23.7-+,  
chr4-56875534-56875666-FLAM-C:Alu:SINE-15.0-+,  
chr2-187524426-187524577-L2a:L2:LINE-30.6-+  
Negative: chr5-121361610-121362462-L1M5:L1:LINE-28.6--,  
chr5-121314389-121315652-L1PA13:L1:LINE-18.2--,

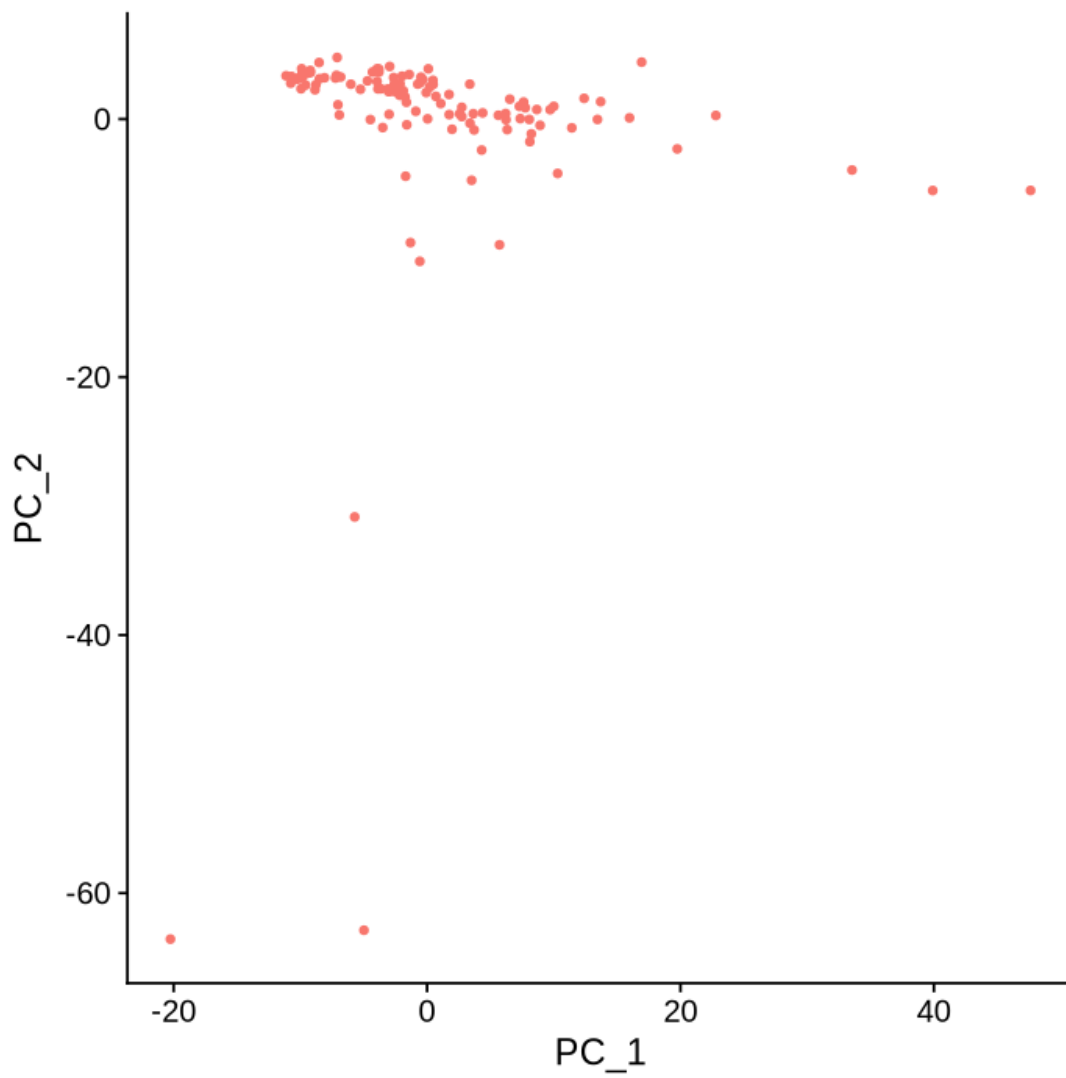
chr5-121362666-121363571-L1MCc:L1:LINE-35.1-+,  
 chr5-121339304-121339825-L1M2:L1:LINE-17.7--,  
 chr7-90691136-90691999-L2a:L2:LINE-26.9-- , chr5-121378192-121378334-MSTB2:ERVL-  
 MaLR:LTR-17.5-+, chrX-78062813-78063242-MLT2A1:ERVL:LTR-8.7--,  
 chr4-22666029-22666158-HAL1M8:L1:LINE-17.7--,  
 chr5-121322741-121324489-L1MEg:L1:LINE-27.9--,  
 chr4-159473812-159475347-L1PA17:L1:LINE-12.6-+  
     chr4-1994283-1994574-AluSz:Alu:SINE-12.3-+ ,  
 chr2-169566168-169566233-AluJr:Alu:SINE-16.7-- ,  
 chr5-121362480-121362663-L1M5:L1:LINE-30.1-- ,  
 chr5-121316112-121317661-L1PA13:L1:LINE-23.5-- ,  
 chr5-121376088-121376213-L1MEf:L1:LINE-29.4-+ ,  
 chrX-78058410-78059111-L1HS:L1:LINE-0.8-+ ,  
 chr1-11092516-11092818-AluSz:Alu:SINE-11.9-- ,  
 chr5-121340412-121341911-L1M2:L1:LINE-17.7-- ,  
 chrX-78058171-78058409-L1HS:L1:LINE-0.4-- ,  
 chrX-113942522-113942816-AluSp:Alu:SINE-12.9--  
     chr9-107607663-107607974-AluSq2:Alu:SINE-11.2-- ,  
 chr17-60238522-60238822-AluSx1:Alu:SINE-10.0-+ ,  
 chr5-133835804-133837870-L1MEg:L1:LINE-24.6-- ,  
 chrX-111706475-111706778-AluJo:Alu:SINE-15.5-+ ,  
 chrX-113924237-113927372-L1MA4:L1:LINE-19.8-- ,  
 chr16-70666994-70667300-AluJo:Alu:SINE-13.0-+ ,  
 chr8-79379077-79379566-L1MEc:L1:LINE-24.7-+ ,  
 chr4-159453760-159454057-AluSg:Alu:SINE-12.1-- ,  
 chrX-111753237-111756423-L1PA7:L1:LINE-7.9-+ ,  
 chrX-113935840-113936232-L2c:L2:LINE-33.9-+  
 PC\_ 4  
 Positive: chr11-120387357-120387430-L2a:L2:LINE-21.1-+ ,  
 chr13-37426758-37427329-L1PB1:L1:LINE-8.9-+ ,  
 chr5-69061835-69062138-AluSp:Alu:SINE-9.9-+ ,  
 chr18-49414994-49415102-AluSx3:Alu:SINE-17.4-- ,  
 chr13-107445948-107446235-AluSc:Alu:SINE-10.1-- ,  
 chr2-151456728-151457033-AluSq2:Alu:SINE-16.0-- ,  
 chr19-45663121-45663420-AluSq2:Alu:SINE-9.7-+ ,  
 chr13-75311682-75312288-L1M5:L1:LINE-33.7-- ,  
 chr15-34917701-34918002-AluJo:Alu:SINE-15.9-+ ,  
 chr3-45458411-45458698-AluSg:Alu:SINE-12.2-+  
     chr1-199888300-199888597-AluSz:Alu:SINE-13.4-+ ,  
 chr2-63973411-63973562-L1MCb:L1:LINE-16.5-- ,  
 chr11-36189204-36189507-AluSz:Alu:SINE-10.5-+ ,  
 chr11-66621375-66621676-AluSq:Alu:SINE-12.2-- ,  
 chr5-99680241-99680525-AluJb:Alu:SINE-19.6-+ ,  
 chr21-37445862-37445991-FLAM-C:Alu:SINE-13.1-+ ,  
 chr16-8017672-8017789-L1MD2:L1:LINE-22.7-+ ,  
 chr12-32810981-32812303-Tigger4:TcMar-Tigger:DNA-12.6-- ,  
 chr17-9315734-9316034-AluSx:Alu:SINE-12.0-+ ,  
 chr8-103949313-103951311-L1M5:L1:LINE-23.3-+

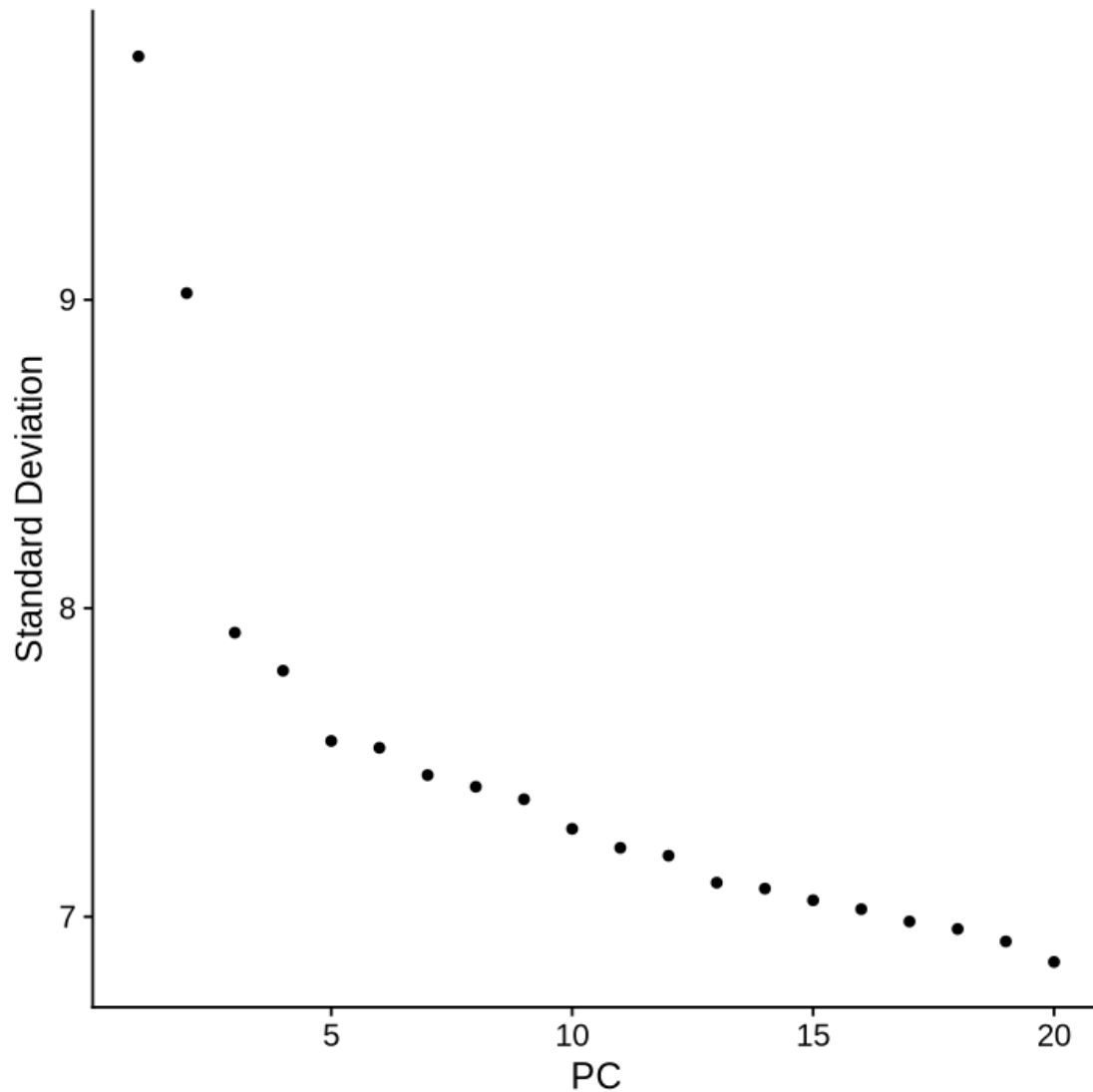
chr10-96150735-96151052-AluSx1:Alu:SINE-11.0--,  
 chr2-31952848-31953087-L1ME4a:L1:LINE-30.3--,  
 chr10-74168949-74169237-AluJb:Alu:SINE-18.0-+, chr11-116303185-116304218-ERVL-E-  
 int:ERVL:LTR-24.4-+, chr1-15193568-15193856-AluSx:Alu:SINE-8.3--,  
 chr9-125567906-125569515-MER41-int:ERV1:LTR-13.4-+,  
 chr14-58229397-58229697-AluY:Alu:SINE-6.0-+,  
 chr2-38347768-38348066-AluSx1:Alu:SINE-13.4--,  
 chr5-98913952-98914269-MLT1J:ERVL-MaLR:LTR-28.0-+,  
 chr11-107340927-107341290-Tigger17c:TcMar-Tigger:DNA-23.2-+  
 Negative: chr12-70505981-70506273-AluJr:Alu:SINE-17.8-+,  
 chr20-18515951-18516252-L2a:L2:LINE-26.6-+,  
 chr10-118660589-118662884-L1PBa:L1:LINE-8.3--,  
 chr2-10350600-10351475-L1MB8:L1:LINE-14.8-+,  
 chr12-112428748-112429046-AluSp:Alu:SINE-7.4--,  
 chr1-145754964-145755248-AluSz:Alu:SINE-11.9-+,  
 chr13-79352619-79352851-MIRc:MIR:SINE-24.7--,  
 chr10-93029899-93030207-AluSp:Alu:SINE-9.1--,  
 chr22-36320957-36321244-AluSx1:Alu:SINE-9.7--,  
 chr17-28895400-28895562-AluYg6:Alu:SINE-12.3--  
 chr14-34600127-34600423-AluSz6:Alu:SINE-15.2--,  
 chr15-34950079-34950387-AluSc:Alu:SINE-9.7--,  
 chr13-46574145-46574347-L1MA2:L1:LINE-8.5-+,  
 chr19-22766950-22767008-L1M4c:L1:LINE-29.3-+,  
 chr14-45234936-45235222-AluSx:Alu:SINE-13.2-+,  
 chr16-71804735-71805012-AluJb:Alu:SINE-15.5-+,  
 chr4-189259692-189260070-MSTA:ERVL-MaLR:LTR-14.9--,  
 chr9-104843488-104843645-MIRb:MIR:SINE-27.4-+,  
 chr11-65409387-65409490-L1MC5a:L1:LINE-28.8--,  
 chr12-12848664-12848842-MSTA:ERVL-MaLR:LTR-12.8-+  
 chr20-43672941-43673244-AluSz:Alu:SINE-12.5--,  
 chr22-17391427-17391614-MIRc:MIR:SINE-27.0-+,  
 chr6-79072942-79073033-MIR:MIR:SINE-23.5--,  
 chr19-21477254-21477533-AluSp:Alu:SINE-10.7-+,  
 chr22-31728331-31728624-AluSx1:Alu:SINE-7.5-+,  
 chr2-30501476-30501688-AluSx3:Alu:SINE-14.1-+,  
 chr1-32654091-32654389-AluJb:Alu:SINE-15.1-+,  
 chr6-37884212-37884522-AluSp:Alu:SINE-9.7-+,  
 chr22-40877927-40878223-AluSq2:Alu:SINE-10.4-+,  
 chr20-13554583-13555101-Tigger1:TcMar-Tigger:DNA-14.7--  
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 Positive: chr11-90658971-90659986-L1MD1:L1:LINE-29.6--,  
 chr6-63566560-63568698-Tigger1:TcMar-Tigger:DNA-12.8--,  
 chr3-130916984-130917322-MER47A:TcMar-Tigger:DNA-13.3-+,  
 chr1-12792582-12792820-AluSz:Alu:SINE-20.1--,  
 chr6-63551080-63551365-AluSx:Alu:SINE-11.5--,  
 chr9-94539183-94539473-AluSx:Alu:SINE-11.3--,  
 chr16-89515718-89516008-AluSx3:Alu:SINE-10.0--,  
 chr9-121323143-121323371-L1MC4a:L1:LINE-16.2--,

chr11-90660560-90661011-MLT2A2:ERVL:LTR-10.0--,  
 chrY-2913363-2915317-L1M1:L1:LINE-19.6--  
     chr2-231088339-231089554-Tigger1:TcMar-Tigger:DNA-11.8--+,  
 chr21-16490494-16490790-AluSx:Alu:SINE-12.1--+,  
 chr7-26270257-26270743-Tigger1:TcMar-Tigger:DNA-11.5--,  
 chr2-65095356-65095662-AluJr:Alu:SINE-16.6--,  
 chr1-113101456-113101653-MER5C1:hAT-Charlie:DNA-20.7--+,  
 chr10-70156118-70156351-L2b:L2:LINE-25.7--,  
 chr6-89083299-89083654-L1PA16:L1:LINE-13.0--,  
 chr18-31934587-31934854-AluSq2:Alu:SINE-8.6--+,  
 chr1-226304917-226305183-AluSz:Alu:SINE-13.5--+,  
 chr1-75383478-75388208-L1PA2:L1:LINE-2.6--  
     chr14-45252199-45252487-L1ME3G:L1:LINE-24.9--+,  
 chr10-102161239-102161531-AluJb:Alu:SINE-21.5--,  
 chr6-24699209-24699509-AluY:Alu:SINE-5.3--,  
 chr2-10440961-10441247-AluJb:Alu:SINE-16.7--,  
 chr8-73536088-73537891-L1MDa:L1:LINE-22.4--+,  
 chr4-115446636-115452847-L1PA6:L1:LINE-7.8--,  
 chr4-128062769-128062922-L1PA4:L1:LINE-2.0--+,  
 chr4-168971584-168973362-L1PB1:L1:LINE-11.3--,  
 chr14-91425985-91426467-THE1B-int:ERVL-MaLR:LTR-10.5--+,  
 chr14-91426468-91426756-AluSz6:Alu:SINE-13.5--  
 Negative: chr16-74790111-74790873-L1MC4:L1:LINE-24.4--,  
 chr2-24608567-24609263-L1MD3:L1:LINE-18.3--,  
 chr11-66808254-66808543-AluJo:Alu:SINE-17.9--+,  
 chr5-121340412-121341911-L1M2:L1:LINE-17.7--,  
 chr5-121362480-121362663-L1M5:L1:LINE-30.1--,  
 chr5-121316112-121317661-L1PA13:L1:LINE-23.5--,  
 chr1-37541720-37542409-MER44C:TcMar-Tigger:DNA-16.9--,  
 chr12-51616934-51618458-L1ME1:L1:LINE-22.5--,  
 chr11-63825946-63826235-AluSz:Alu:SINE-11.7--,  
 chr5-121322741-121324489-L1MEg:L1:LINE-27.9--  
     chr10-21539973-21540108-AluJb:Alu:SINE-13.5--+,  
 chr16-68031463-68031662-MER65D:ERV1:LTR-16.9--+,  
 chr5-121378192-121378334-MSTB2:ERVL-MaLR:LTR-17.5--+,  
 chr13-110707365-110707659-AluSx:Alu:SINE-10.2--+,  
 chr4-22666029-22666158-HAL1M8:L1:LINE-17.7--,  
 chr16-74777161-74777348-MER96B:hAT-Tip100:DNA-14.9--+,  
 chr10-26841281-26842489-L1MA9:L1:LINE-18.7--,  
 chr20-503463-503895-MER119:hAT-Charlie:DNA-22.6--+,  
 chr5-121339304-121339825-L1M2:L1:LINE-17.7--,  
 chr12-12848664-12848842-MSTA:ERVL-MaLR:LTR-12.8--  
     chr5-72127501-72127625-L1PA8:L1:LINE-8.4--+,  
 chr2-203482181-203482480-AluSx1:Alu:SINE-11.3--,  
 chr8-47280691-47280836-AluJr:Alu:SINE-17.1--,  
 chr11-65409387-65409490-L1MC5a:L1:LINE-28.8--,  
 chr7-152051230-152051364-L1MC5a:L1:LINE-24.4--,  
 chrX-78062813-78063242-MLT2A1:ERVL:LTR-8.7--,  
 chr11-46427727-46428031-AluSx1:Alu:SINE-8.5--+,



chr15-80984292-80984425-FLAM-C:Alu:SINE-20.9-+,  
chr18-31543584-31543884-AluJb:Alu:SINE-15.9-+,  
chr5-121362666-121363571-L1Mcc:L1:LINE-35.1-+





```
[24]: objTE_SoloTE <- FindNeighbors(objTE_SoloTE, dims = 1:13, k.param = 20)
objTE_SoloTE <- FindClusters(objTE_SoloTE, resolution = 1, algorithm=4)
objTE_SoloTE <- RunUMAP(objTE_SoloTE, dims = 1:13)
DimPlot(objTE_SoloTE, reduction = "umap")
```

Computing nearest neighbor graph

Computing SNN

Warning message in RunLeiden(object = object, method = method, partition.type = "RBConfigurationVertexPartition", :  
 "`random.seed` must be greater than 0 for leiden clustering, resetting  
 `random.seed` to 1."



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15:55:29 Writing NN index file to temp file  
/mnt/TEdataStorage\_2T/tmp/RtmpjoyMJm/file376f761b7f8fa0

15:55:29 Searching Annoy index using 1 thread, search\_k = 3000

15:55:29 Annoy recall = 100%

15:55:34 Commencing smooth kNN distance calibration using 1 thread  
with target n\_neighbors = 30

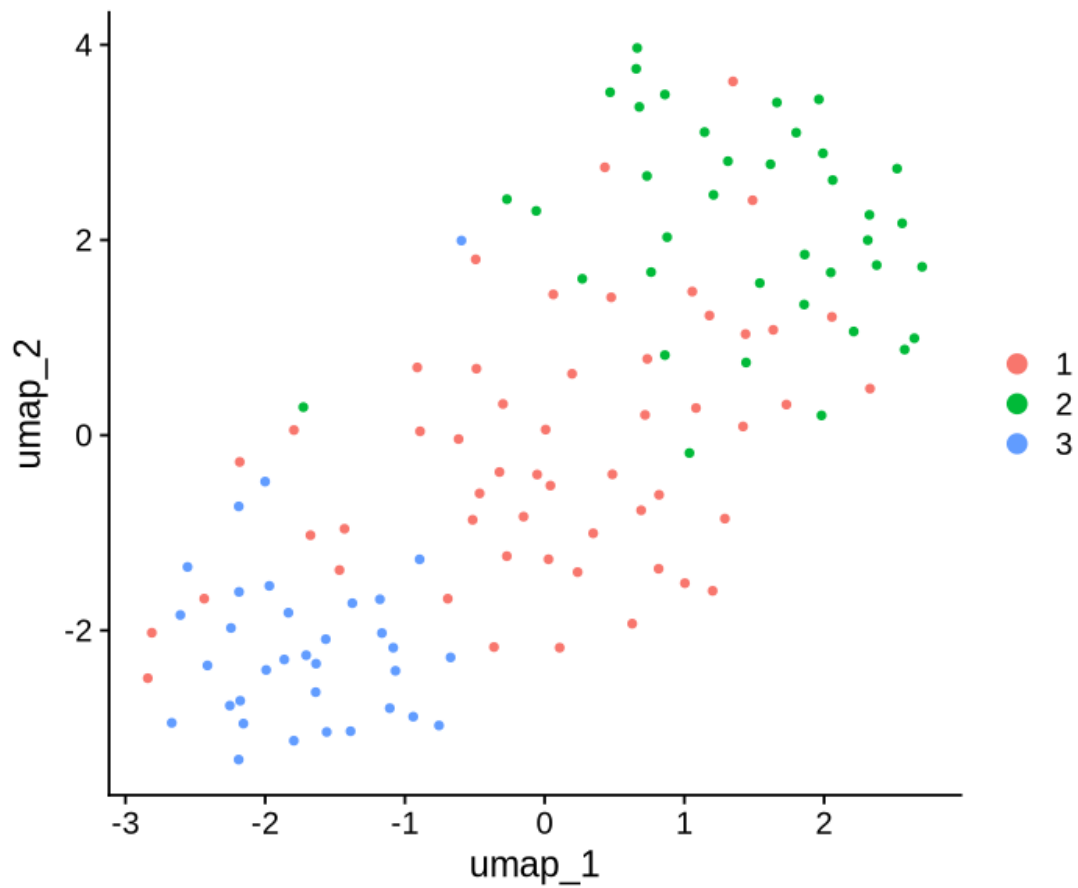
15:55:34 8 smooth knn distance failures

15:55:44 Initializing from normalized Laplacian + noise (using RSpectra)

15:55:44 Commencing optimization for 500 epochs, with 4830 positive edges

15:55:44 Using rng type: pcg

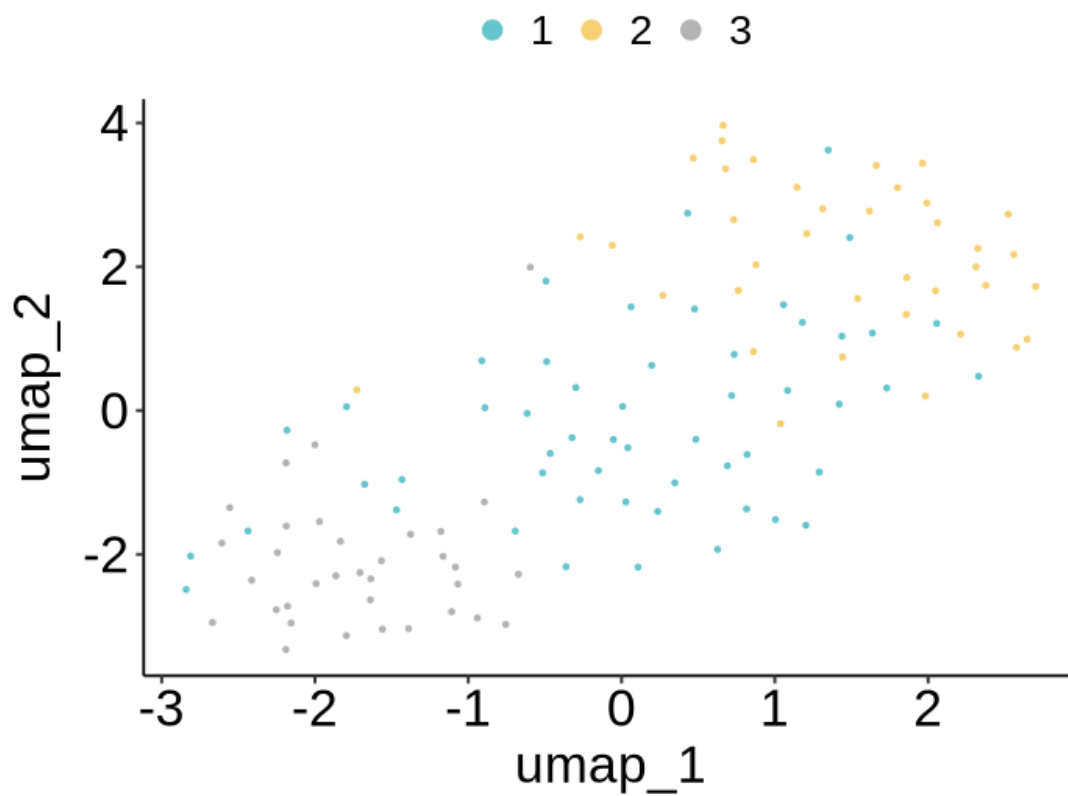
15:55:49 Optimization finished



```
[25]: options(repr.plot.width=6, repr.plot.height=5)

DimPlot(objTE_SoloTE, reduction = "umap", group.by = "seurat_clusters",
        shuffle=T, pt.size = 0.5) +
  theme_pubr() +
  scale_color_carto_d(palette = "Pastel") +
  theme(text=element_text(size=20))
ggsave(paste0("figures_", dataset_id, "/umap_locus_clusters.pdf"), device = "pdf", width=6, height=5)
```

## seurat\_clusters



```
[26]: # # Save objects
saveRDS(objGenes_SoloTE, paste0("data_", dataset_id, "/soloTE_", dataset_id, "_",
  ↪ "_GENE_seuratObj.RDS"))
saveRDS(objTE_SoloTE, paste0("data_", dataset_id, "/soloTE_", dataset_id, "_",
  ↪ "_seuratObj.RDS"))
```